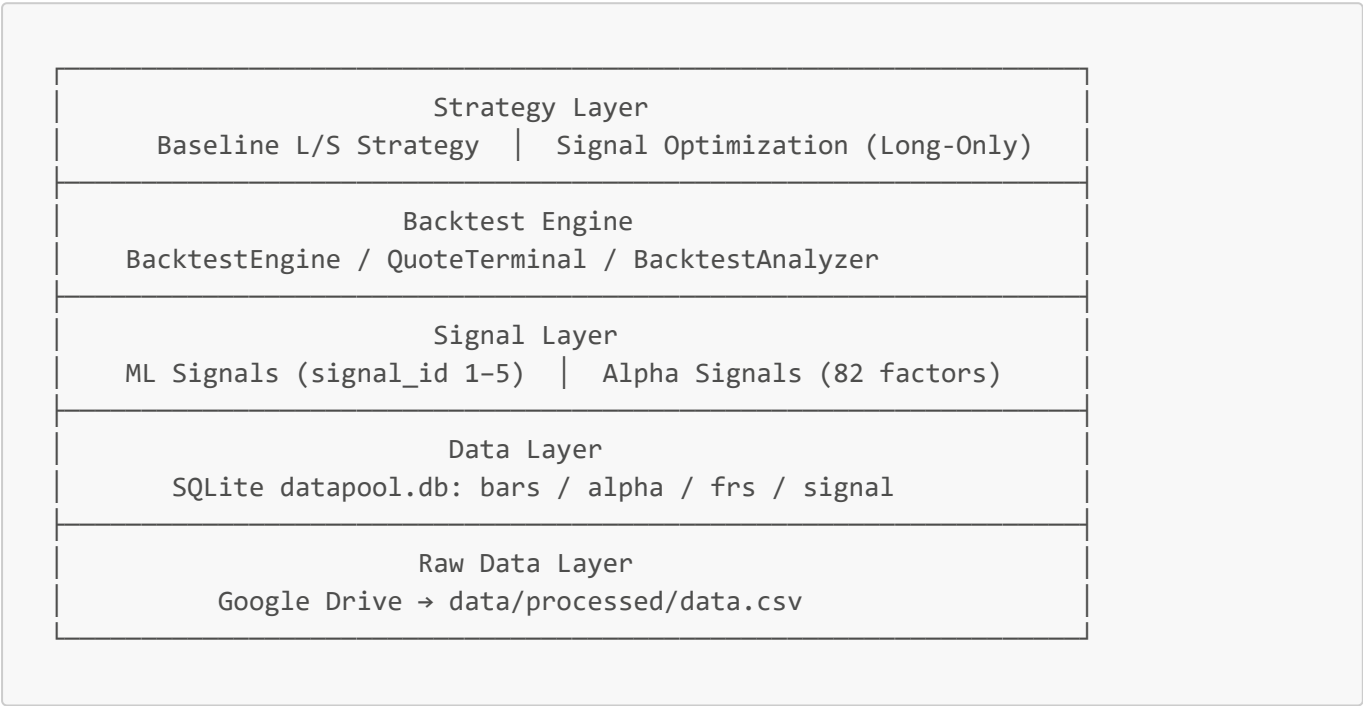


QuantLab Project — Architecture & Progress Report

Date: 2026-05-13 | **Branch:** `feature-infra`

1. Project Architecture Overview

The system is built in five layers, bottom-up:



2. Layer-by-Layer Description

2.1 Data Infrastructure

Docs: `docs/01 Data Infra.md`, `docs/02 Data Instruction.md`

Universe: 11 SPDR Sector ETFs (XLB, XLC, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLRE, XLY) + SPY/SPX/VIX/US10Y benchmark series. Data is ingested from Google Drive into `data/processed/data.csv`, then pushed into `datapool.db` (SQLite) via `scripts/dataset_builder.ipynb`.

Database schema (key tables):

Table	Content	Frequency
<code>daily_bar</code>	OHLCV + TRI, all 15 assets	Daily
<code>weekly_bar</code>	Wednesday-close resample, all 15 assets	Weekly
<code>weekly_alpha</code>	82 alpha factor values, ETFs only	Weekly
<code>weekly_frs</code>	FRS1/2/3 forward return scores, ETFs only	Weekly
<code>weekly_signal</code>	ML / alpha / composite signals, ETFs only	Weekly

FRS definitions (ML model prediction targets / labels):

Code	Name	Formula
FRS1	4-week total return	$(P_4 - P_0) / P_0$
FRS2	Sharpe proxy	$\text{avg}(r_1 \dots r_4) / \text{std}(r_1 \dots r_4)$
FRS3	Vol-penalised return	$\text{FRS1} - 2.0 \times \text{std}(r_1 \dots r_4)$

Developer workflow for extending metrics: add a decorated function to `frs_metrics.py` / `alpha_metrics.py` / `signal_metrics.py`, then re-run the corresponding `save_*` function. Registry rows are append-only; fact tables are refreshed per run.

2.2 Alpha Pool — 101 Formulaic Alphas

Docs: `docs/05 Alpha.md` | **Code:** `src/QuantLab/alpha/alpha_metrics.py`

Implemented from Kakushadze (2015) *101 Formulaic Alphas*:

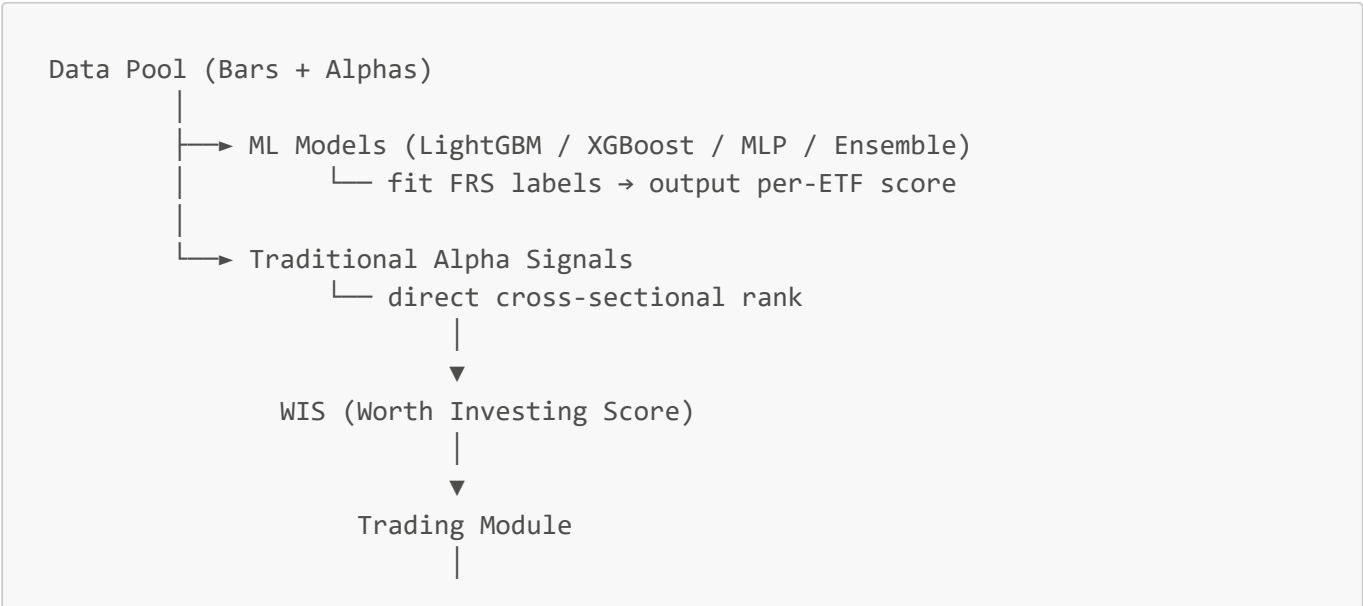
Group	Count	Note
Group A — Fully implementable	52	All required fields available
Group B — vwap approximated	30	$\text{vwap} \approx (O+H+L+C)/4$
Not implementable	19	Requires <code>IndNeutralize</code> or market-cap data

Top alphas by IC (Mean IC > 0.02): Alpha#50 (0.042), #3 (0.034), #41 (0.034), #24 (0.033), #98 (0.031).

The project also defines **custom alphas (alpha_id 108–136)**, including `alpha_110` (12-week cumulative return), which is hard-wired into the risk module as an absolute-momentum filter for short positions.

2.3 ML Architecture

Docs: `docs/03 ML architecture.md`



▼

Backtest Engine

Five ML signals are currently trained and stored in `weekly_signal` (signal_id 1–5):

signal_id	Model	Label
1	LightGBM	FRS3
2	Ensemble (Rank Avg)	FRS1
3	XGBoost	FRS3
4	PCA + Ridge	FRS3
5	MLP	FRS2

2.4 Backtest Engine

Docs: docs/06 Backtest Engine.md | **Code:** src/QuantLab/backtest/

Four-module design with strict separation of concerns:

Module	Role	Key API
BacktestEngine	Advances simulation clock, coordinates all modules, applies fills, calls evaluation	<code>engine.run()</code> , <code>engine.evaluate()</code>
QuoteTerminal	Single source of truth for time + data (prevents look-ahead bias)	<code>terminal.at(day)</code> , <code>terminal.signals()</code> , <code>terminal.etfs()</code>
Signal / Strategy / Risk	Signal → ranking → position intent → risk filter	<code>analyze()</code> , <code>on_ranking()</code> , <code>on_action()</code>
BacktestAnalyzer	Performance metrics + multi-panel dashboard (NAV vs SPY)	outputs: <code>.json</code> , <code>.md</code> , <code>.csv</code> , <code>.png</code>

Key contract: all historical queries are cut off at `terminal.day` (inclusive); Signal/Strategy/Risk share the same terminal via `bind(terminal)`.

Outputs per run:

- `*_metrics.json` / `*_metrics.md`
- `*_value_history.csv` / `*_holding_history.csv`
- `*_all_in_one_panel.png` (Panel #1: NAV vs SPY)

2.5 Signal Optimization Framework (Long-Only)

Docs: docs/07 Signal Optimization.md | **Code:** backtests/signal_optimization/

Strategy: `SiganlOptimizationStrategy` — softmax weight allocator, fully invested long-only.

$$w_i = \frac{\exp(s_i)}{\sum_j \exp(s_j)}$$

Objective: $\mathcal{L} = \text{Sharpe} - \lambda \cdot \text{Turnover}_{\text{ann}}$, where $\lambda = 0.1$.

Four-step optimization path:

Step	What is optimized	Method
0	Equal-weight baseline (mandatory floor)	—
1	Single signal or small parameter set	IC pre-screen + grid sweep
2	Linear mix $\sum_k \alpha_k g^{(k)}$	Bayesian optimization + walk-forward CV
3	Neural network (per-ETF logits)	Gradient descent via differentiable backtest

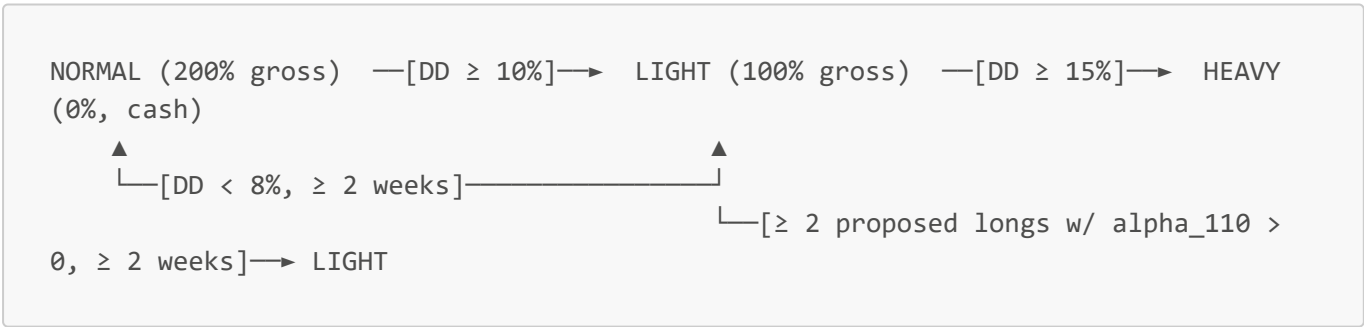
2.6 Baseline Long-Short Strategy

Docs: `docs/08 baseline strategy.md` | **Code:** `src/QuantLab/backtest/strategy/baseline.py`, `risk/baseline.py`

Design: Market-neutral long-short portfolio, weekly rebalancing. Long top-3 / Short bottom-3 ETFs, target weight $\pm 33.3\%$ each, gross exposure 200%, net exposure $\sim 0\%$.

Short filter: a short position is only opened if `alpha_110` (12-week momentum) < 0 . This prevents shorting ETFs in upward trends.

Risk state machine (`BaselineRisk`):



Rank stickiness: existing holdings are retained if their current rank stays within `n_long + 2` (long) or `n_total - n_short + 1 - 2` (short), avoiding whipsaw from minor rank fluctuations.

3. Backtest Results Summary

Test period: **2025-01-01** → **2026-01-31** (57 weekly periods) | Initial NAV: 10,000 | Zero fee, zero slippage.

3.1 Equal-Weight Baseline (Step 0 — mandatory floor)

Metric	Value
Sharpe Ratio	1.305

Metric	Value
Annual Return	13.98%
Annual Volatility	10.46%
Max Drawdown	-10.64%
Turnover (ann.)	28.24%
Excess Return vs SPY	-3.08%
CAPM Alpha vs SPY	2.37%

This is the benchmark that every signal must beat.

3.2 Signal Optimization — Alpha Screening (40 alphas tested)

All 40 alphas tested with `SignalOptimizationStrategy` (softmax, long-only). Key results:

alpha_id	Sharpe	Ann. Return	Max DD	Turnover Ann.
19 ★	1.968	23.33%	-7.26%	2,752%
24	1.758	24.60%	-11.32%	2,598%
23	1.665	18.78%	-10.66%	1,772%
31	1.561	16.57%	-9.30%	2,164%
Equal-Weight (\$0)	1.305	13.98%	-10.64%	28%

Best alpha: Alpha#19 (Sharpe = 1.968, Ann. Return = 23.33%, CAPM Alpha = 11.35% vs SPY)

Alpha#19 formula: `-sign((close - delay(close,7)) + delta(close,7)) × (1 + rank(1 + sum(returns,250)))` — a long-horizon momentum reversal signal with a 250-day warm-up.

Note: several low-turnover alphas (32, 57, 61, 64, 66, 72, 83) all produce identical metrics (Sharpe 1.305), indicating they produce constant or near-constant cross-sectional scores — effectively equal-weight.

3.3 Signal Optimization — ML Signal Screening (Step 1)

All 5 ML signals tested under the same framework ($\lambda = 0.1$):

Signal	Sharpe	Ann. Return	Max DD	Turnover Ann.	L = Sharpe – $\lambda \cdot \text{TO}$	Δ vs \$0
Ensemble_RankAvg_frs1 ★	1.506	16.76%	-10.27%	652.67%	1.506	+0.201
MLP_frs2	1.329	14.06%	-10.26%	577.08%	1.329	+0.024
Equal-Weight (\$0)	1.305	13.98%	-10.64%	28.24%	1.305	0.000

Signal	Sharpe	Ann. Return	Max DD	Turnover Ann.	L = Sharpe – λ ·TO	Δ vs \$0
XGBoost_frs3	1.304	13.97%	-10.62%	41.62%	1.304	-0.001
PCA_Ridge_frs3	1.304	13.96%	-10.44%	35.20%	1.304	-0.001
LightGBM_frs3	1.300	13.92%	-10.64%	41.53%	1.300	-0.005

Best ML signal: Ensemble_RankAvg_frs1 (Signal 2) — Sharpe = 1.506, beats the equal-weight baseline by +0.201 Sharpe.

The three FRS3-based models (LightGBM, XGBoost, PCA+Ridge) barely outperform equal-weight. The ensemble averaging FRS1 rank predictions is the only ML signal that beats the baseline significantly.

3.4 Baseline Long-Short — Alpha Screening

Same 40 alphas tested with **BaselineStrategy** (long 3 / short 3, market-neutral):

alpha_id	Sharpe	Ann. Return	Ann. Vol	Max DD	Turnover Ann.
23 ★	2.081	23.00%	10.21%	-5.85%	4,802%
57	1.528	17.72%	11.08%	-7.40%	5,425%
37	1.478	16.98%	11.03%	-5.78%	3,965%
10	1.230	13.37%	10.67%	-6.90%	4,910%
51	1.098	9.95%	9.01%	-5.84%	4,288%

Many alphas yield negative Sharpe in the long-short setting (e.g., alpha#14 = -1.35, alpha#44 = -1.50, alpha#116 = -1.66), meaning the short side often hurts performance. **Alpha#23** stands out as the best signal for the L/S framework.

Best: Alpha#23 — Sharpe = **2.081**, Ann. Return = 23.00%, Max DD = **-5.85%**, CAPM Alpha vs SPY = **17.33%**, Beta = 0.234 (very low market exposure).

Alpha#23 formula: $(\text{sum}(\text{high}, 20) / 20 < \text{high}) ? -\text{delta}(\text{high}, 2) : 0$ — a high-breakout momentum-reversal signal.

3.5 Baseline Long-Short — ML Signal Screening

Signal	Sharpe	Ann. Return	Ann. Vol	Max DD	Turnover Ann.
Ensemble_RankAvg_frs1 ★	0.806	9.62%	12.32%	-4.59%	4,544%
PCA_Ridge_frs3	-0.083	-1.53%	11.20%	-7.98%	4,092%
MLP_frs2	-0.388	-4.12%	9.67%	-10.40%	4,978%
XGBoost_frs3	-0.468	-3.38%	6.85%	-10.76%	2,766%

Signal	Sharpe	Ann. Return	Ann. Vol	Max DD	Turnover Ann.
LightGBM_frs3	-1.681	-9.85%	6.05%	-13.37%	2,636%

ML signals perform significantly worse in the long-short context. Only Ensemble_RankAvg_frs1 achieves a positive Sharpe (0.806), while the others lose money. This suggests the ML signals have directional information useful for long selection, but the short side predictions are unreliable.

4. Cross-Strategy Performance Comparison

Strategy	Signal	Sharpe	Ann. Return	Max DD	Beta vs SPY
Baseline L/S	Alpha#23	2.081	23.00%	-5.85%	0.234
Sig. Opt. (Long-Only)	Alpha#19	1.968	23.33%	-7.26%	0.613
Sig. Opt. (Long-Only)	Signal 2 (Ensemble)	1.506	16.76%	-10.27%	0.691
Equal-Weight Baseline	—	1.305	13.98%	-10.64%	0.674
Baseline L/S	Signal 2 (Ensemble)	0.806	9.62%	-4.59%	0.275

Key observations:

1. **Baseline L/S with Alpha#23 is the top performer** overall (Sharpe 2.081, CAPM Alpha 17.33%), with near-zero market beta — a strong market-neutral result.
2. **Long-only Alpha#19** achieves higher raw return (23.33%) but carries more market exposure and higher turnover (2,752% ann.).
3. **ML signals underperform alpha signals in the L/S framework** — only the ensemble model survives with a positive Sharpe.
4. **Equal-weight is a surprisingly hard benchmark to beat** for ML signals in the long-only setting.

5. Current Progress Status

Done

Module	Status	Note
Data pipeline & <code>datapool.db</code>	✓ Complete	End-to-end pipeline from CSV to all DB tables
Alpha pool (82 + custom)	✓ Complete	IC computed, custom alphas (108–136) including <code>alpha_110</code>
ML signals (signal_id 1–5)	✓ Complete	Trained and stored in <code>weekly_signal</code>
Backtest engine	✓ Complete	<code>QuoteTerminal</code> / <code>BacktestEngine</code> / <code>BacktestAnalyzer</code>

Module	Status	Note
Signal Opt. Step 0 (equal-weight)	 Complete	Sharpe 1.305 — mandatory benchmark established
Signal Opt. alpha screening	 Complete	40 alphas; best: alpha#19 (Sharpe 1.968)
Signal Opt. Step 1 (ML signals)	 Complete	5 signals; best: Ensemble_frs1 (Sharpe 1.506)
Baseline L/S — alpha screening	 Complete	40 alphas; best: alpha#23 (Sharpe 2.081)
Baseline L/S — ML signal screening	 Complete	5 signals; best: Ensemble_frs1 (Sharpe 0.806)
Baseline Strategy + Risk framework	 Complete	3-state risk machine, rank stickiness, short momentum filter

In Progress / Pending

Module	Status	Note
Signal Opt. Step 2 (multi-signal Bayesian)	 Next	Linear mix $\sum \alpha_k g^{\{k\}}$ + walk-forward CV
Signal Opt. Step 3 (neural network)	 Planned	Differentiable backtest path
Baseline parameter tuning	 Planned	Optimal <code>stickiness_threshold</code> , borrow cost estimate, liquidity filter
Transaction cost sensitivity study	 Planned	Sweep <code>long_cost</code> over {0, 5, 10, 20} bps
Final performance report	 Planned	Full comparison table with in-sample / out-of-sample split

6. Directory Map

LAB/	
— docs/	# Design documents (01–08)
— src/QuantLab/	
— alpha/	# Alpha registration & compute
— frs/	# FRS registration & compute
— signal/	# Signal registration (incl. ML sub-module)
— backtest/	
— engine.py	# Main simulation loop
— quote_terminal.py	# Unified data access (anti-look-ahead)
— analyzer.py	# Metrics & visualization
— signal/	# AlphaBacktestSignal, MLBacktestSignal,

```

etc.
|  |  | strategy/                # SigalOptimizationStrategy,
BaselineStrategy
|  |  | risk/                    # BaselineRisk, DebugRisk
|  |  | utils/                   # db / data_loader / load_ml_input / config
|  |  | backtests/
|  |  |   signal_optimization/
|  |  |   | smoketest/           # Step 0 equal-weight [DONE]
|  |  |   | alphas/              # 40-alpha screening [DONE] → best:
alpha#19
|  |  |   | step1/                # ML signal Step 1 [DONE] → best: Signal 2
|  |  |   | baseline/
|  |  |   | alpha/               # L/S alpha screening [DONE] → best:
alpha#23
|  |  |   | ml_signal/            # L/S ML screening [DONE] → best: Signal 2
|  |  |   | scripts/
|  |  |   | dataset_builder.ipynb # DB build orchestration

```

7. Key Design Decisions

Decision	Rationale
Wed-close signal / Thu-close execution	Eliminates look-ahead bias; compresses weekend gap to a single overnight
QuoteTerminal as the single time truth	All modules share the same data cut-off via <code>bind(terminal)</code> , preventing date drift
alpha_110 hard-wired short filter	Prevents shorting ETFs in upward 12-week trends; non-negotiable risk rule
Softmax allocation (no hard top-k)	Preserves gradient flow for Step 3 neural network training
Soft floor $\epsilon = 1e-6$	Numerical continuity for differentiable backtest paths
Equal-weight as mandatory Sharpe floor	Forces signals to prove cross-sectional information content, not just market beta